

Simulating Voices: Does Instruction Tuning Make a Simulated Crowd Hyper-Accurate?

A replication of the wisdom-of-crowds Turing Experiment with an open base / instruct LLM pair

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1. Hypothesis

Context. LLMs serve as *synthetic participants* in behavioural and opinion research [Aher; Santurkar]. This only works if simulated people show *human-like variation*.

Prior work: the hyper-accuracy distortion

Asked “How many bones does an adult human have?”, humans answer with median 190 (IQR 108); GPT-4 answers 206, the exact value, with IQR 0 [Aher, Tab. 3–4]. Aher et al. suspect alignment as the cause, but could only test closed models whose training details are not released [Aher, p. 25].

Research question & hypothesis

RQ Does instruction tuning *alone* make a simulated crowd give exact answers without human-like spread?

H1 With the *same prompt*, an instruction-tuned model gives (a) **more exact answers** and (b) a **smaller spread** than its base model.

Contribution: a test with an open model pair in which only post-training differs.

2. Methodology

Design

Models: Qwen2.5-0.5B vs. Qwen2.5-0.5B-Instruct [Qwen Team].

Data: 10 questions with true values; human median/IQR for Q1–5 [Aher, Tab. 3].

Crowd: 50 simulated people (25 surnames × Mr./Ms.) per question.

Prompt & sampling as Aher et al.: Fig. 14, temperature 1, top- p 1; invalid answers re-sampled.

10 questions × 50 simulated people → Aher et al. prompt

base
same prompt

instruct
same prompt

instruct
chat template

Measures: exact answers (bootstrap 95 % CI); spread = IQR / true value. **Tests** vs. base: Fisher exact (a), Wilcoxon over 10 questions (b); Bonferroni $\alpha=0.0125$

Code & all 1,500 raw outputs:

github.com/ssowrabha249-coder/

Simulating-Voices-and-Fabricating-Realities--Poster

3. Results

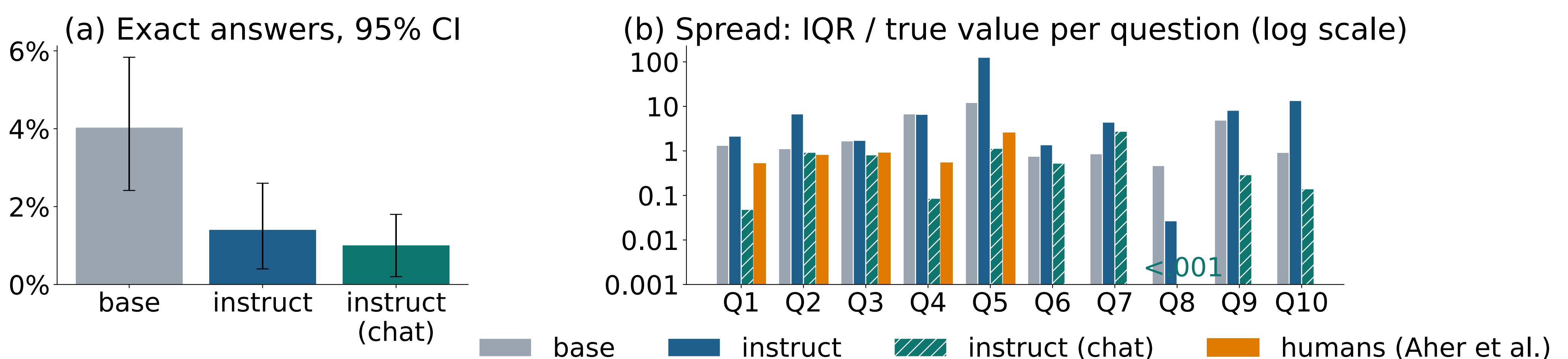


Figure. (a) Pooled share of exactly correct answers, 95 % bootstrap CI. (b) IQR / true value per question (log scale); human values from Aher et al., Tab. 3 (Q1–Q5 only). Q8 (speed of light, 299,792,458): the ratio is tiny for all conditions and does not mean accurate answers.

Summary per condition

	exact (95 % CI)	spread	spread Q1–5
base	4.0 % (2.4 %–5.8 %)	1.23	1.68
instruct	1.4 % (0.4 %–2.6 %)	5.53	6.64
instr. (chat)	1.0 % (0.2 %–1.8 %)	0.41	0.82
humans	–	–	0.81

spread = median over questions of IQR / true value.

Interpretation

hyper-accuracy may need alignment *plus* the knowledge of larger models; Aher et al.’s smallest models did not show it either [Aher, Tab. 3].

Discussion vs. hypothesis

H1(a) not supported, opposite direction: both instruct conditions give *fewer* exact answers than base (Fisher $p = 0.011$, $p = 0.002$; significant after correction). No condition has IQR = 0 on any question (GPT-4: all 10 [Aher, Tab. 4]).

H1(b) not supported: same prompt $\Rightarrow$ *more* spread on 8 of 10 questions ($p = 0.020$, n.s. after correction).

Unexpected: the chat format narrows the spread (9/10 questions, $p = 0.037$, n.s.); Q1–5 close to humans (0.82 vs. 0.81) but around *wrong* values (median: 24 bones).

Limitations

One model size (0.5B), one prompt wording, 10 questions; human values taken from Aher et al., not collected; spread ratio is misleading for Q8.

Take-home: instruction tuning alone did **not** make a small simulated crowd hyper-accurate. Check spread and correctness **separately**; next: repeat across model sizes.